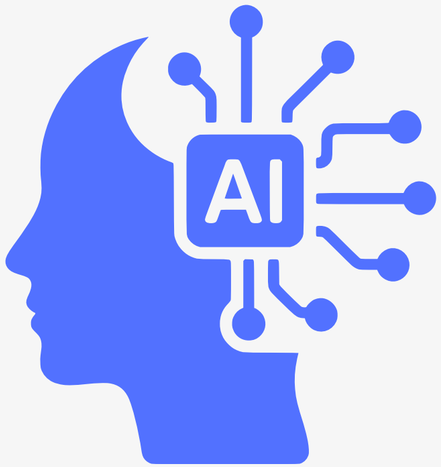
</>TechZeen



AGENTIC AI SYSTEMS

WHAT IF AI COULD DECIDE WHAT TO DO NEXT, USE TOOLS, CHECK RESULTS, AND KEEP WORKING UNTIL A TASK IS COMPLETE?

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WHAT IS AN AI WORKFLOW?

- An AI workflow follows a predefined sequence of steps to complete a task.
- Each step has a specific job, and the next step usually runs according to predefined rules.
- Example: Receive an email → extract its important information → summarize it → save the summary.

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WHAT IS AN AI AGENT?

- An AI agent can observe a task, decide what action to take, and use tools to make progress toward a goal.
- It can examine results and choose its next action instead of following only one fixed sequence.
- Example: Ask an agent to research the best laptops under $500. It can search, compare specifications, check prices, and prepare a recommendation.

DETERMINISTIC FLOWS

- A deterministic flow follows predefined rules and steps.
- Given the same inputs and conditions, it is designed to produce the same sequence of actions.
- Example: If an order is paid, send a confirmation email. If payment fails, show an error.

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AUTONOMOUS LOOPS

- An autonomous loop allows an agent to observe results, decide what to do next, and repeat actions until a goal or stopping condition is reached.
- The agent may change its approach when an action fails or new information appears.
- Example: An agent tries to find a working API endpoint, examines the error, and adjusts its next attempt within approved limits.

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ORCHESTRATION THINKING

- Orchestration means coordinating multiple steps, tools, or AI components to complete a larger task.
- A workflow can orchestrate fixed steps, while an agent can make decisions within an orchestrated system.
- Example: A research system searches websites, summarizes findings, checks sources, and prepares a final report.

FUNCTION CALLING

- Function calling allows an AI model to request a specific function with structured arguments.
- The model typically selects a tool and provides its inputs, but the application executes the actual function.
- Example: The AI requests get_weather(city="Karachi") to retrieve the weather.

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TOOL EXECUTION

- Tool execution is the process of running the requested function or tool and obtaining its result.
- The result is returned to the AI, which can use it to answer the user or decide what to do next.
- Example: The AI requests a weather API call, the application runs it, and the AI explains the returned forecast.

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BROWSER TOOLS

- Browser tools allow an AI system to interact with web pages or retrieve information from websites.
- Depending on their permissions, they may search, navigate pages, extract information, or interact with page elements.
- Example: An agent searches three laptop websites, collects prices, and compares the available options.

API TOOLS

- API tools let an AI application communicate with external services through defined endpoints.
- They can retrieve data or perform actions, depending on the API's permissions.
- Example: An AI assistant calls a task-management API to create a new task.

OS INTERACTION

- Operating system tools allow an AI system to interact with approved files, processes, or terminal commands.
- These capabilities must be restricted because unrestricted access could delete files, expose secrets, or run unsafe commands.
- Example: A coding agent reads project files and runs approved tests to check whether code works.